THE LEAST CERTIFICATE COORDINATE FOR THE ERDŐS–STRAUS EQUATION AT A PRIME: A TWO-SHIFT REDUCTION, PARTIAL SIEVE RESULTS, AND COVERING OBSTRUCTIONS

ANONYMOUS

ABSTRACT. For an integer $n \equiv 1 \pmod 4$ call a triple (t, u, v) of positive integers with $tuv - u - v = n$ a *word* of n, and let $\mathcal{C}_n$ be the set of positive integers that are $\equiv 0 \pmod 4$ or $\equiv -n \pmod 8$. A word with a coordinate in $\mathcal{C}_n$ yields an explicit solution of $4/n = 1/x + 1/y + 1/z$, and $c \in \mathcal{C}_n$ is a coordinate of a word if and only if $n + c$ or $cn + 1$ has a divisor $\equiv -1 \pmod c$. We study the least such coordinate $\mathrm{HC}(p)$ at primes p. It equals 3 for $p \equiv 5 \pmod 8$ and is at most 8 unless p lies in one of Mordell's six classes modulo 840. For $p \equiv 1 \pmod 8$ we show: $\mathrm{HC}(p) > 4$ holds exactly when $p + 4$ and $4p + 1$ have only prime factors $\equiv 1 \pmod 4$; every prime $p = 5m^2 + 1$ with $4 \mid m$ has $\mathrm{HC}(p) \geq 7$; under the Elliott–Halberstam conjecture at some level $1 - \eta_0 < 1$ there are $\gg x/(\log x)^2$ primes $p \leq x$ with $\mathrm{HC}(p) > 4$, which is the true order; and unconditionally $\gg x/(\log x)^2$ primes $p \leq x$, $p \equiv 1 \pmod 8$, have $p + 4$ composed of primes $\equiv 1 \pmod 4$ and $4p + 1$ free of primes $\equiv 3 \pmod 4$ below x^δ for an absolute $\delta > 0$. We show that no finite family of fixed certificates covers the primes in a class $p \equiv 1 \pmod{840}$, $p \equiv R \pmod \ell$ with R a small quadratic non-residue modulo a prime $\ell \geq 11$; such classes contain no squares, so this obstruction is not the classical one at squares. We also bound the reach of Dickson-type constructions of primes with large HC. A computation gives $\mathrm{HC}(p) \leq 64$ for every prime $p \equiv 1 \pmod 4$ below 10^{12}. The Erdős–Straus conjecture remains open; nothing here proves that $\mathrm{HC}(p)$ is finite for all large p, nor that it is unbounded.

1. INTRODUCTION

The Erdős–Straus conjecture asserts that for every integer $n \geq 2$

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \tag{1}$$

has a solution in positive integers. It is open. It suffices to treat primes, and classical polynomial identities settle every prime outside the residue classes $1, 121, 169, 289, 361, 529 \pmod{840}$ (Mordell [10]; see [3, p. 8]). Finite families of such identities cannot settle any class consisting of squares (Yamamoto [16], Schinzel [13]; see [3, Prop. 1.6 and the discussion before Prop. 1.9]). The equation has been verified numerically for $n \leq 10^{17}$ [12], and a recent preprint

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reports $n \le 10^{18}$ [9]. On average there are many solutions: Elsholtz and Tao [3, Thm. 1.1] show $N(\log N)^2 \ll \sum_{p \le N} f(p) \ll N(\log N)^2 \log\log N$ for the number $f(p)$ of solutions.

This note isolates one family of explicit solutions and studies the size of its smallest parameter. For $n \equiv 1 \pmod 4$ a *word* of n is a triple $(t,u,v) \in \mathbb{N}^3$ with

$$tuv - u - v = n,$$

and we put

$$\mathcal{C}_n := \{x \in \mathbb{N} : \ x \equiv 0 \pmod 4 \text{ or } x \equiv -n \pmod 8\}.$$

If some coordinate of a word lies in $\mathcal{C}_n$, then one of five explicit identities solves (1) (Proposition 2.2). The coordinate $c \in \mathcal{C}_n$ occurs in a word exactly when $n + c$ or $cn + 1$ has a positive divisor $d \equiv -1 \pmod c$ (Lemma 2.3); we call such a pair (c,d) a *certificate* and define

$$\mathrm{HC}(n) := \min\{c : \ (c,d) \text{ is a certificate for } n\} \in \mathbb{N} \cup \{\infty\}.$$

So $\mathrm{HC}(p) < \infty$ implies that (1) is solvable for $n = p$. We do not know whether the converse holds. The coordinates divisible by 4 are the "Type A" and "Type B" solutions of López [8], who conjectures that every prime has one (Remark 2.5).

$\mathrm{HC}(p) = 3$ for $p \equiv 5 \pmod 8$, so the interesting primes are $p \equiv 1 \pmod 8$, where $\mathcal{C}_p = \{4, 7, 8, 12, 15, 16, 20, 23, \dots\}$. A uniform bound $\mathrm{HC}(p) \le F(p)$ for all large p, for any function F, would prove the Erdős–Straus conjecture for all large primes. We prove no such bound. Our results concern the following questions: how small is HC on explicit classes, how often and by which mechanism can it be large, and what can a covering argument achieve.

Results. Throughout, $\mathcal{Q}$ is the set of primes $\equiv 3 \pmod 4$ and $\mathcal{S}$ is the set of positive integers all of whose prime factors are $\equiv 1 \pmod 4$.

(1) *Basic structure (§2).* Every certificate divisor d for n has Jacobi symbol $(d/n) = -1$ (Lemma 2.7). Hence $\mathrm{HC}(m^2) = \infty$ for every odd m, and certificates supported on primes dividing $2m$ never occur when $n \equiv 1 \pmod{8m}$.

(2) *Explicit classes and density (§§3, 4).* $\mathrm{HC}(p) \le 8$ for every prime $p > 7$, $p \equiv 1 \pmod 4$, outside Mordell's six classes modulo 840 (Proposition 3.1), and $\#\{p \le x : \ p \equiv 1\ (8),\ \mathrm{HC}(p) > 4\} \ll x/(\log x)^2$ (Proposition 4.8).

(3) *The first threshold (§4).* For $p \equiv 1 \pmod 8$, $\mathrm{HC}(p) > 4$ if and only if $p + 4 \in \mathcal{S}$ and $4p + 1 \in \mathcal{S}$ (Lemma 4.1). So "$\mathrm{HC}(p) > 4$ for infinitely many p" is the two-shift prime problem

$$\text{infinitely many primes } p \equiv 1 \pmod 8 \text{ with } p + 4 \in \mathcal{S} \text{ and } 4p + 1 \in \mathcal{S}, \quad (2)$$

an analogue of the hyperbolic prime number theorem of Friedlander and Iwaniec [5] (Remark 4.2). We prove:

- *Theorem Q.* Every prime $p = 5m^2 + 1$ with $4 \mid m$ has $\mathrm{HC}(p) \geq 7$. Hence (2) follows from Bunyakovsky's conjecture for the single polynomial $720k^2 + 1$.
- *Theorem E.* There is an absolute $\eta_0 > 0$ such that the Elliott–Halberstam hypothesis $\mathrm{EH}(1 - \eta_0)$ implies $\#\{p \leq x : \ p \equiv 1\ (8),\ p{+}4 \in \mathcal{S},\ 4p{+}1 \in \mathcal{S}\} \gg x/(\log x)^2$. With Proposition 4.8 this is the true order.
- *Theorem U* (unconditional). There are absolute $\delta > 0$ and $c > 0$ with

$$\#\{p \leq x : \ p \equiv 1\ (8),\ p{+}4 \in \mathcal{S},\ 4p{+}1 \text{ has no prime factor in } \mathcal{Q} \text{ below } x^{\delta}\} \geq c\,\frac{x}{(\log x)^2}$$

  for large x. For these p, $4p + 1$ has a bounded number of prime factors in $\mathcal{Q}$; (2) needs that number to be 0.

(4) *Dickson-type constructions (§5).* Under Dickson's conjecture HC is unbounded (Proposition 5.1). Theorem W shows that a fixed polynomial family of this kind can certify $\mathrm{HC}(p) > X$ only for $X < (4 + o(1)) \log p$, except possibly at its least member.

(5) *Coverings (§6). Theorem N*: let $\ell \geq 11$ be prime and R a quadratic non-residue modulo ℓ with $1 \leq R < (\ell - 1)/3$. For every C and every finite set $\mathcal{D}$ of integers there are infinitely many primes $p \equiv 1 \pmod{840}$, $p \equiv R \pmod{\ell}$, having no certificate (c, d) with $c \leq C$ and $d \in \mathcal{D}$. The class contains no squares.

(6) *Computation (§7).* $\mathrm{HC}(p) \leq 64$ for all 18 803 924 340 primes $p \equiv 1 \pmod 4$ below 10^{12}; the value 64 occurs only at $p = 21\,841$.

What is not proved. We do not prove the Erdős–Straus conjecture, nor $\mathrm{HC}(p) < \infty$ for all large p, nor any bound $\mathrm{HC}(p) \leq F(p)$ for all large p. We do not prove (2): Theorems Q and E are conditional and Theorem U leaves a bounded number of prime factors from $\mathcal{Q}$ in $4p + 1$. Theorem W limits one construction only; it does not bound HC. Theorem N excludes fixed finite certificate families only; arguments whose moduli grow with p are not excluded. The computation is a finite verification (§7 describes the checks); the Erdős–Straus conjecture itself is verified much further [12, 9].

Relation to the literature. The solutions in Proposition 2.2(c) are of Type I or Type II in the sense of [3]; for instance the words with $4 \mid t$ are the case $c = 1$ of the characterisation $4abcd = a + b + nc$ of Type II solvability in [3, Prop. 2.7]. The divisor criterion of Lemma 2.3 for $c \equiv 0 \pmod 4$ is López's Type A condition $p \equiv -4d \pmod{4dn - 1}$ and its level analogue [8]. Theorem E adapts the method of Friedlander–Iwaniec [5, Thm. 4, §6], who treat $\sum_{n \leq x} \Lambda(n) r(n-2) r(n+2)$ under a level-of-distribution hypothesis. Theorem U adapts the vector sieve of Nath and Xie [11], who obtain primes $p = m^2 + n^2 + 1$ with $\Omega(p + 2)$ bounded. We did not find the statements of Theorems Q, E, U or N in the literature; our search was limited to arXiv and web-indexed sources (MathSciNet and zbMATH were not searched). A

related one-parameter statistic, the least $a \equiv 3 \pmod 4$ for which $(p+a)/4$ is a denominator of a solution, is studied in the companion note [15].

Conventions. p and q denote primes, ℓ usually denotes a prime, $(\cdot/\cdot)$ is the Jacobi symbol, and $\Omega_{\mathcal{Q}}(n)$ is the number of prime factors of n in $\mathcal{Q}$, counted with multiplicity. Constants called absolute do not depend on any parameter.

2. Words, certificates and the statistic HC

Lemma 2.1. *Let $n \in \mathbb{Z}$ and $t,u,v \in \mathbb{N}$, and put $A = tu-1$, $A^* = tv-1$.*

(a) *The following are equivalent: $tuv-u-v = n$; $vA = n+u$; $uA^* = n+v$; $AA^* = tn+1$.*

(b) *If $n \geq 1$ and (t,u,v) is a word of n, then $A, A^* \geq 1$ and $t,u,v \leq n+2$.*

Proof. (a) $vA - u = tuv - u - v$, symmetrically for uA^*, and $AA^* = t(tuv - u - v) + 1$ with $t \neq 0$. (b) If $tu = 1$ then $tuv - u - v = -1$, so $A \geq 1$; likewise $A^* \geq 1$. $t = (n+u+v)/(uv) \leq n+2$. By symmetry let $u \leq v$. Then $v(tu-1) = n+u \leq n+v$, so $v(tu-2) \leq n$; if $tu \geq 3$ this gives $v \leq n$, and if $tu = 2$ then $v = n+u \leq n+2$. □

Proposition 2.2. *Let $n \equiv 1 \pmod 4$, $n \geq 5$, and let (t,u,v) be a word of n, with $A = tu-1$, $A^* = tv-1$.*

(a) *If one of t,u,v lies in $\mathcal{C}_n$, then 4 divides one of the five numbers t,u,v,A,A^*.*

(b) *Conversely, if 4 divides one of t,u,v,A,A^*, then one of t,u,v lies in $\mathcal{C}_n$.*

(c) *If 4 divides one of t,u,v,A,A^*, then (1) has the following solution:*

$$\begin{aligned} t = 4s: &\quad \frac{4}{n} = \frac{1}{suv} + \frac{1}{nsv} + \frac{1}{nsu};\\ u = 4r: &\quad \frac{4}{n} = \frac{1}{rA^*} + \frac{1}{tnr} + \frac{1}{tnrA^*};\\ A = 4a: &\quad \frac{4}{n} = \frac{1}{av} + \frac{1}{avA^*} + \frac{1}{naA^*}; \end{aligned}$$

and symmetrically for $v = 4r$ and $A^* = 4a$ (exchange $u \leftrightarrow v$, $A \leftrightarrow A^*$).

Proof. (c) Using Lemma 2.1(a): for $t = 4s$, $\frac{1}{suv} + \frac{1}{nsv} + \frac{1}{nsu} = \frac{n+u+v}{nsuv} = \frac{tuv}{nsuv} = \frac{4}{n}$. For $u = 4r$, the sum is $\frac{tn+A^*+1}{tnrA^*} = \frac{t(n+v)}{tnrA^*} = \frac{uA^*}{nrA^*} = \frac{4}{n}$, since $A^*+1 = tv$. For $A = 4a$, the sum is $\frac{nA^*+n+v}{navA^*} = \frac{ntv+v}{navA^*} = \frac{v\,AA^*}{navA^*} = \frac{4}{n}$, since $AA^* = tn+1$.

(a) If the coordinate in $\mathcal{C}_n$ is divisible by 4 there is nothing to prove. Suppose $u \equiv -n \pmod 8$ with u odd. Then $vA = n+u \equiv 0 \pmod 8$, so v or A is divisible by 4. The case of v is symmetric. If $t \equiv -n \pmod 8$ with t odd, then $AA^* = tn+1 \equiv 1-n^2 \equiv 0 \pmod 8$, since n is odd; so $4 \mid A$ or $4 \mid A^*$.

(b) We may assume that none of t, u, v is divisible by 4 and, by symmetry, that $4 \mid A = tu - 1$. Then t and u are odd and $n = vA - u$. If $8 \mid A$, or if $A \equiv 4 \pmod 8$ and v is even, then $n \equiv -u \pmod 8$, so $u \in \mathcal{C}_n$. Otherwise $A \equiv 4 \pmod 8$ and v is odd, so $n \equiv 4 - u \pmod 8$ and $tu \equiv 5 \pmod 8$. As $n \equiv 1 \pmod 4$ we get $u \equiv 3 \pmod 4$. If $u \equiv 3 \pmod 8$ then $t \equiv 7$ and $n \equiv 1 \pmod 8$; if $u \equiv 7 \pmod 8$ then $t \equiv 3$ and $n \equiv 5 \pmod 8$. In both cases $t \equiv -n \pmod 8$, so $t \in \mathcal{C}_n$. □

So a word of n has a coordinate in $\mathcal{C}_n$ if and only if one of its five quantities t, u, v, A, A^* is divisible by 4, and each such word gives an explicit solution of (1). All denominators in (c) are positive.

Lemma 2.3. *Let $n \geq 1$ and $c \geq 2$.*

(U) *c is the second (equivalently, by the symmetry $(t, u, v) \mapsto (t, v, u)$, the third) coordinate of a word of n if and only if $n + c$ has a positive divisor $d \equiv -1 \pmod c$. Given d, the word is $((d+1)/c,\ c,\ (n+c)/d)$.*

(T) *c is the first coordinate of a word of n if and only if $cn+1$ has a positive divisor $d \equiv -1 \pmod c$. Given d, the word is $(c,\ (d+1)/c,\ (e+1)/c)$ with $e = (cn + 1)/d$.*

Proof. (U) If (t, c, v) is a word, $d = tc - 1 \geq 1$ divides $n + c$ by Lemma 2.1(a), and $d \equiv -1 \pmod c$. Conversely $d + 1$ is a positive multiple of c, so $t = (d + 1)/c \in \mathbb{N}$, and $v = (n + c)/d$ satisfies $v(tc - 1) = n + c$. (T) If (c, u, v) is a word, $d = cu - 1$ divides $cn + 1$ by Lemma 2.1(a). Conversely, $de = cn + 1 \equiv 1 \pmod c$ and $d \equiv -1$ give $e \equiv -1 \pmod c$, so $u = (d + 1)/c$ and $v = (e + 1)/c$ are positive integers with $(cu - 1)(cv - 1) = cn + 1$. □

Definition 2.4. Let $n \equiv 1 \pmod 4$. A *certificate* for n is a pair (c, d) with $c \in \mathcal{C}_n$ and d a positive divisor of $n + c$ or of $cn + 1$ satisfying $d \equiv -1 \pmod c$; we call d a *certificate divisor*. The *U-line* and the *T-line* at c are the integers $n + c$ and $cn + 1$. We put $\mathrm{HC}(n) := \min\{c : (c, d) \text{ a certificate}\}$, with $\min \emptyset = \infty$.

Since $1, 2 \notin \mathcal{C}_n$, Lemma 2.3 shows that $\mathrm{HC}(n)$ is the least element of $\mathcal{C}_n$ that is a coordinate of some word of n, and by Proposition 2.2 $\mathrm{HC}(n) < \infty$ implies that (1) is solvable.

Remark 2.5. A word with $4 \mid u$ corresponds to a divisor $d = 4kn' - 1$ of $p + 4k$, i.e. $p \equiv -4k \pmod{4kn' - 1}$; a word with $4 \mid t$, $t = 4k$, $u = n'$ corresponds to $p \equiv -n' \pmod{4kn' - 1}$. These are López's Type A and Type B solutions [8, Thms. 4 and 7], and his Conjecture 1 is equivalent to: every prime has a word with a coordinate divisible by 4. Our $\mathcal{C}_p$ also contains the odd classes $\equiv -p \pmod 8$.

Proposition 2.6. *Let $p \equiv 1 \pmod 4$ be prime.*

(a) *$\mathrm{HC}(p) \geq 3$, and $\mathrm{HC}(p) = 3$ if and only if $p \equiv 5 \pmod 8$; then $(1, 3, (p + 3)/2)$ is a word.*

(b) *If* $p \equiv 1 \pmod 8$, *then* $\mathrm{HC}(p) \geq 4$, *and* $\mathrm{HC}(p) = 4$ *if and only if* $(p+4)(4p+1)$ *has a prime factor* $\equiv 3 \pmod 4$. *If* $\mathrm{HC}(p) > 4$ *then* $\mathrm{HC}(p) \geq 7$.

(c) *If* $\mathrm{HC}(p) < \infty$ *then* $\mathrm{HC}(p) \leq p - 1$.

Proof. (a) $1, 2 \notin \mathcal{C}_p$ because $-p$ is odd and $\not\equiv 1 \pmod 8$. $3 \in \mathcal{C}_p$ iff $p \equiv 5 \pmod 8$, and then $d = 2 \mid p+3$ with $2 \equiv -1 \pmod 3$. (b) For $p \equiv 1 \pmod 8$, $\mathcal{C}_p = 4\mathbb{N} \cup (7 + 8\mathbb{Z}_{\geq 0})$, so its two least elements are 4 and 7. The numbers $p+4$ and $4p+1$ are odd; an odd number has a divisor $\equiv 3 \pmod 4$ iff it has a prime factor $\equiv 3 \pmod 4$, because products of numbers $\equiv 1 \pmod 4$ are $\equiv 1 \pmod 4$. (c) By Lemma 2.1(b) every coordinate is $\leq p+2$, and $p, p+1, p+2 \notin \mathcal{C}_p$: they are $\equiv 1, 2, 3 \pmod 4$, and the odd ones are not $\equiv -p \pmod 8$ since $2p$, $2p+2 \not\equiv 0 \pmod 8$. □

2.1. **Reciprocity.** The following lemma is the only arithmetic property of p beyond congruences that enters the covering results; it is the certificate form of the classical obstruction at squares.

Lemma 2.7. *Let* $n \equiv 1 \pmod 4$ *be a positive integer and* (c, d) *a certificate for* n. *Then* $\gcd(d, n) = 1$ *and* $(d/n) = -1$.

Proof. Since $d \equiv -1 \pmod c$, $\gcd(c, d) = 1$. Write $d = 2^a d'$ with d' odd. If $d \mid n + c$ then $n \equiv -c \pmod{d'}$; if $d \mid cn + 1$ then $n \equiv -\bar{c} \pmod{d'}$ with $c\bar{c} \equiv 1 \pmod{d'}$. As $(\bar{c}/d') = (c/d')$, in both cases $(n/d') = (-c/d')$, which is ± 1. By Jacobi reciprocity and $n \equiv 1 \pmod 4$, $(d'/n) = (n/d')$; this is nonzero, so $\gcd(n, d') = 1$, and $\gcd(n, 2) = 1$. Hence

$$\left(\frac{d}{n}\right) = \left(\frac{2}{n}\right)^a \left(\frac{-c}{d'}\right). \tag{3}$$

Case $4 \mid c$. Then $d \equiv -1 \pmod 4$, so $a = 0$, $d' = d \equiv 3 \pmod 4$ and $(-1/d) = -1$. Write $c = 4k$, $k = 2^b k'$ with k' odd. If $b \geq 1$ then $8 \mid c \mid d + 1$, so $d \equiv 7 \pmod 8$ and $(2/d) = 1$. By reciprocity, as $(d-1)/2$ is odd, $(k'/d) = (d/k')(-1)^{(k'-1)/2}$, and $d \equiv -1 \pmod{k'}$ gives $(d/k') = (-1/k') = (-1)^{(k'-1)/2}$; so $(k'/d) = 1$. Hence $(c/d) = 1$ and $(-c/d) = -1$, and (3) gives $(d/n) = -1$.

Case c *odd.* Then $c \equiv -n \pmod 8$, so $c \equiv 3 \pmod 4$. By reciprocity $(c/d') = (d'/c)(-1)^{(d'-1)/2} = (d'/c)(-1/d')$, so $(-c/d') = (d'/c)$. From $2^a d' \equiv -1 \pmod c$ we get $(2/c)^a (d'/c) = (-1/c) = -1$. Since $c \equiv -n \pmod 8$ and $(2/m)$ depends only on $\pm m \bmod 8$, $(2/c) = (2/n)$. So (3) gives $(d/n) = (2/n)^a \cdot (-(2/n)^a) = -1$. □

For a prime p the lemma says that every certificate divisor is a quadratic non-residue modulo p; in particular it has a prime factor that is a non-residue.

Corollary 2.8. (a) *If* m *is odd, then* m^2 *has no certificate:* $\mathrm{HC}(m^2) = \infty$.

(b) *Let* $m \geq 1$ *and* $n \equiv 1 \pmod{8m}$. *Then no certificate divisor of* n *has all its prime factors among the primes dividing* $2m$.

(c) *For every finite set* $\mathcal{D} \subset \mathbb{N}$ *there are infinitely many primes* p *having no certificate* (c, d) *with* $d \in \mathcal{D}$.

Proof. (a) $(d/m^2) = (d/m)^2 \in \{0, 1\}$. (b) $(2/n) = 1$ as $n \equiv 1 \pmod 8$, and for an odd prime $\ell \mid m$, $(\ell/n) = (n/\ell) = (1/\ell) = 1$. So such a d has $(d/n) = 1$, contrary to Lemma 2.7. (c) Apply (b) with m the product of the elements of $\mathcal{D}$, and Dirichlet's theorem. □

Part (a) is the certificate form of the vanishing of Type I and Type II solutions at odd squares [3, Prop. 1.6]. It implies that any proof that $\mathrm{HC}(p) < \infty$ must use a property of p that fails at odd squares.

3. Explicit classes

Proposition 3.1. *Let* $p > 7$ *be a prime with* $p \equiv 1 \pmod 8$. *Then* p *has the following words:*

condition	*word* (t, u, v)	*consequence*
$p \equiv 2 \pmod 3$	$(1, 4, (p+4)/3)$	$\mathrm{HC}(p) = 4$
$p \equiv 3 \pmod 7$	$(2, 4, (p+4)/7)$	$\mathrm{HC}(p) = 4$
$p \equiv 5 \pmod 7$	$(4, 2, ((4p+1)/7+1)/4)$	$\mathrm{HC}(p) = 4$
$p \equiv 3 \pmod 5$	$(3, 7, (p+7)/20)$	$\mathrm{HC}(p) \le 7$
$p \equiv 2 \pmod 5$	$(7, 3, ((7p+1)/20+1)/7)$	$\mathrm{HC}(p) \le 7$
$p \equiv 6 \pmod 7$	$(1, 8, (p+8)/7)$	$\mathrm{HC}(p) \le 8$

Consequently $\mathrm{HC}(p) \le 8$ *for every prime* $p > 7$ *with* $p \equiv 1 \pmod 4$ *and* $p \not\equiv 1, 121, 169, 289, 361, 529 \pmod{840}$.

Proof. In each row the third coordinate is a positive integer and $tuv - u - v = p$: for $(1, 4, v)$, $3v - 4 = p$ and $3 \mid p + 4$; for $(2, 4, v)$, $7v - 4 = p$; for $(1, 8, v)$, $7v - 8 = p$; for $(3, 7, v)$, $20v - 7 = p$, and $20 \mid p + 7$ because $p \equiv 1 \pmod 4$ and $p \equiv 3 \pmod 5$. The rows $(4, 2, \cdot)$ and $(7, 3, \cdot)$ are Lemma 2.3(T) with $(c, d) = (4, 7)$ and $(7, 20)$: $7 \mid 4p + 1$ iff $p \equiv 5 \pmod 7$, and $20 \mid 7p + 1$ iff $p \equiv 17 \pmod{20}$. The coordinates $4, 8$ and 7 lie in $\mathcal{C}_p$, and $\mathrm{HC}(p) \ge 4$ by Proposition 2.6. The conditions in the table say that p is a quadratic non-residue modulo 3, 5 or 7. The primes $p \equiv 1 \pmod 8$ that are residues modulo 3, 5 and 7 are those with $p \equiv 1 \pmod 3$, $p \equiv \pm 1 \pmod 5$ and $p \equiv 1, 2, 4 \pmod 7$, i.e. the six classes listed. For $p \equiv 5 \pmod 8$ we have $\mathrm{HC}(p) = 3$. □

The six classes are the classes of $1, 11^2, 13^2, 17^2, 19^2, 23^2$, i.e. the squares in $(\mathbb{Z}/840\mathbb{Z})^\times$, as Corollary 2.8(b) requires.

4. The first threshold: $\mathrm{HC}(p) > 4$

4.1. Reduction to a two-shift problem.

Lemma 4.1. *Let* $p \equiv 1 \pmod 8$ *be prime. Then* $\mathrm{HC}(p) > 4$ *if and only if* $p + 4 \in \mathcal{S}$ *and* $4p + 1 \in \mathcal{S}$, *and in that case* $\mathrm{HC}(p) \ge 7$. *In particular, for every* $X \in \{4, 5, 6\}$, *there are infinitely many primes* $p \equiv 1 \pmod 8$ *with*

$\mathrm{HC}(p) > X$ *if and only if* (2) *holds; and* (2) *is necessary for* $\mathrm{HC}(p) > X$ *infinitely often, for every* $X \geq 4$.

Proof. Proposition 2.6(b): $p+4$ and $4p+1$ are odd, and an odd number lies in $\mathcal{S}$ iff it has no prime factor $\equiv 3 \pmod 4$. □

Remark 4.2 (A determinant-15 shell)**.** An odd integer $N \geq 1$ lies in $\mathcal{S}$ if and only if $N = a^2 + b^2$ with $\gcd(a,b) = 1$. Let $\Lambda_4 = \{M = \left(\begin{smallmatrix} P & -R \\ T & Q \end{smallmatrix}\right) \in M_2(\mathbb{Z}) : P+Q \equiv R+T \equiv 0 \pmod 4\}$. The map $(x,y,u,v) \mapsto (P,Q,R,T) = (2x-u,\ 2x+u,\ 2y+v,\ 2y-v)$ is a bijection from $\{x^2+y^2 = p+4,\ u^2+v^2 = 4p+1\}$ onto $\{M \in \Lambda_4 :\ \det M = 15,\ \|M\|^2 = 16p+34\}$, with inverse $x = (P+Q)/4$, $y = (R+T)/4$, $u = (Q-P)/2$, $v = (R-T)/2$; this follows from $\|M\|^2 \pm 2\det M = (P \pm Q)^2 + (R \pm T)^2$. Thus (2) asks for primes p at which the determinant-15 shell of height $16p+34$ has a point with both pairs (x,y) and (u,v) primitive. Friedlander and Iwaniec [5, Thms. 3–4] prove the corresponding lower bound $\sum_{n \leq x} \Lambda(n) r(n-2) r(n+2) \gg x$ for the determinant-1 shell, conditionally on a level of distribution $\theta < 1$ sufficiently close to 1 (their hypothesis $A(\theta)$); the upper bound is unconditional. Theorem E below is the analogue of their conditional lower bound for (2); note that $r(N) > 0$ is weaker than $N \in \mathcal{S}$ (e.g. $45 = 3^2 + 6^2$).

Theorem Q. *Let* $m \equiv 0 \pmod 4$ *and suppose that* $p = 5m^2 + 1$ *is prime. Then* $p \equiv 1 \pmod 8$, $12 \mid m$, *and* $\mathrm{HC}(p) \geq 7$. *Consequently, if the polynomial* $720k^2 + 1$ *takes infinitely many prime values (Bunyakovsky's conjecture for this polynomial), then* (2) *holds.*

Proof. We have $p + 4 = 5(m^2+1)$ and $4p+1 = 5((2m)^2+1)$. For even m the numbers m^2+1 and $4m^2+1$ are odd, and an odd prime ℓ dividing a^2+1 has $(-1/\ell) = 1$, i.e. $\ell \equiv 1 \pmod 4$. As $5 \equiv 1 \pmod 4$, both $p+4$ and $4p+1$ lie in $\mathcal{S}$. From $16 \mid 5m^2$ we get $p \equiv 1 \pmod{16}$. If $3 \nmid m$ then $5m^2 + 1 \equiv 0 \pmod 3$, which is impossible for a prime $p > 3$; so $12 \mid m$ and $p = 720k^2 + 1$ with $m = 12k$. Lemma 4.1 gives $\mathrm{HC}(p) \geq 7$. The polynomial $720k^2+1$ is irreducible and takes the value 1 at $k = 0$, so it has no fixed prime divisor. □

On the Bateman–Horn conjecture, Theorem Q gives $\asymp \sqrt{x}/\log x$ primes $p \leq x$ with $\mathrm{HC}(p) \geq 7$, fewer than the order $x/(\log x)^2$ of Theorem E and Proposition 4.8. The mechanism is the one that kills all lines at odd squares: $5p = (5m)^2 + 5$ is a near-square.

4.2. **Sieve inputs.** For $y \geq 2$ and $q \geq 1$ put $E(y;q) := \max_{(c,q)=1} |\pi(y;q,c) - \mathrm{li}(y)/\varphi(q)|$.

Hypothesis 4.3 (EH(θ))**.** For a fixed $\theta \in (0,1)$: for every $A > 0$, $\sum_{q \leq x^\theta} E(x;q) \ll_{A,\theta} x(\log x)^{-A}$.

The Bombieri–Vinogradov theorem [1], [4, Thm. 3.4] gives, for every $A > 0$, some $B > 0$ with $\sum_{q \leq x^{1/2}(\log x)^{-B}} E(x;q) \ll_A x(\log x)^{-A}$. Ford's

version of EH(θ) [4, (7.6)] has an additional maximum over $y \le x$ and implies Hypothesis 4.3; the Elliott–Halberstam conjecture [4, Conj. 7.3] asserts it for every $\theta < 1$.

Lemma 4.4. *Let $k \ge 1$ and $Q \le x$, and suppose $\sum_{q\le Q} E(x;q) \ll_A x(\log x)^{-A}$ for every A. Then $\sum_{q\le Q} k^{\omega(q)} E(x;q) \ll_{A,k} x(\log x)^{-A}$ for every A.*

Proof. For $q \le x$, $E(x;q) \le x/q + 1 + \mathrm{li}(x)/\varphi(q) \ll x/\varphi(q)$, and $\sum_{q\le x} k^{2\omega(q)}/\varphi(q) \le \prod_{\ell\le x}(1 + k^2\ell/(\ell-1)^2) \ll_k (\log x)^{k^2}$. By Cauchy–Schwarz, $\big(\sum k^{\omega(q)}E\big)^2 \le \big(\sum E\big)\big(\sum k^{2\omega(q)}E\big) \ll x(\log x)^{-A'} \cdot x(\log x)^{k^2}$ with A' arbitrary. □

We use the sieve language of [4, §1]: a finite set $\mathcal{A}$ with subsets $\mathcal{A}_\ell$ indexed by primes, $\mathcal{A}_d = \bigcap_{\ell\mid d}\mathcal{A}_\ell$, $S(\mathcal{A},z) = \#(\mathcal{A} \setminus \bigcup_{\ell\le z}\mathcal{A}_\ell)$, a number X and a multiplicative g with $0 \le g(\ell) < 1$, remainders $r_d = \#\mathcal{A}_d - Xg(d)$ and $V(z) = \prod_{\ell\le z}(1-g(\ell))$.

Cited theorem 4.5 (Upper-bound sieve; [4, Thm. 2.4])**.** Assume that for some $\kappa \le \kappa_0$ and $B \le B_0$,

$$\prod_{y\le\ell\le w}(1-g(\ell))^{-1} \le \Big(\frac{\log w}{\log y}\Big)^{\kappa}\exp\Big(\frac{B}{\log y}\Big) \qquad (2 \le y \le w \le z). \tag{4}$$

Then $S(\mathcal{A},z) \le O_{\kappa_0,B_0}(XV(z)) + \sum_{d\le z}\mu^2(d)|r_d|$.

The lower bounds use the beta sieve of Rosser and Iwaniec in the form quoted by Nath and Xie [11, §3.2–3.3] from Friedlander–Iwaniec [6, Thm. 11.12]. We have not inspected [6] directly. For a set of primes $\mathcal{P}$ write $P(z) = \prod_{\ell\in\mathcal{P},\ell<z}\ell$, and let g be multiplicative, supported on divisors of $P(z)$, with $0 \le g(\ell) < 1$ and

$$\prod_{\ell\in\mathcal{P},\ w\le\ell<z}(1-g(\ell))^{-1} \le \Big(\frac{\log z}{\log w}\Big)^{\kappa}\Big(1+\frac{L}{\log w}\Big) \qquad (2 \le w < z) \tag{5}$$

[11, (3.4)]. The beta-sieve weights $\lambda^{\pm}(d)$ of level D and dimension κ take values in $\{0,\mu(d)\}$, are supported on $d \mid P(z)$, $d \le D$, and satisfy $\sum_{d\mid n}\lambda^-(d) \le [n=1] \le \sum_{d\mid n}\lambda^+(d)$ for $n \mid P(z)$. Put $V^{\pm}(D,z) = \sum_d \lambda^{\pm}(d)g(d)$.

Cited theorem 4.6 (Beta sieve; [11, Lemma 3.6, Remark 3.7, §5])**.** Assume (5), let $\kappa \in \{1/2, 1\}$ and $s = \log D/\log z$. Then $V^+(D,z) \le (F_\kappa(s) + O((\log D)^{-1/6}))V(z)$ for $s \ge \beta_\kappa - 1$ and $V^-(D,z) \ge (f_\kappa(s) + O((\log D)^{-1/6}))V(z)$ for $s \ge \beta_\kappa$, where $V(z) = \prod_{\ell\mid P(z)}(1-g(\ell))$, $\beta_{1/2} = 1$, $\beta_1 = 2$, and

$$F_1(s) = \frac{2e^\gamma}{s}\ (1 \le s \le 3), \qquad f_1(s) = \frac{2e^\gamma\log(s-1)}{s}\ (2 \le s \le 4);$$

$$F_{1/2}(s) = 2\Big(\frac{e^\gamma}{\pi s}\Big)^{1/2}\ (0 < s \le 2), \qquad f_{1/2}(s) = \Big(\frac{e^\gamma}{\pi s}\Big)^{1/2} 2\operatorname{arccosh}\sqrt{s}\ (1 \le s \le 3),$$

and for $s > 2$, resp. $s > 1$, $(\sqrt{s}F_{1/2}(s))' = f_{1/2}(s-1)/(2\sqrt{s})$ and $(\sqrt{s}f_{1/2}(s))' = F_{1/2}(s-1)/(2\sqrt{s})$.

(In [11, §5] the function $f_{1/2}$ is written as $(e^\gamma/\pi s)^{1/2}\log(1+2(s-1)+2\sqrt{s(s-1)})$, which equals the expression above.) Since $\lambda^- \in \{0,\mu\}$ and $\#\mathcal{A}_d = Xg(d) + r_d$, the lower-bound property gives the standard inequality

$$S(\mathcal{A},\mathcal{P},z) \ge \sum_d \lambda^-(d)\#\mathcal{A}_d \ge XV^-(D,z) - \sum_{d|P(z),\, d\le D} |r_d|. \tag{6}$$

Lemma 4.7. *Write $F = F_{1/2}$, $f = f_{1/2}$. Then: (i) $0 \le f(s) \le 1 \le F(s)$ for $s \ge 1$; (ii) F is nonincreasing on $(0,\infty)$; (iii) $F(s) - f(s) \le 1.51\, s^{-1/2}$ for $s \ge 2$; (iv) $f(s) \ge F(1)\sqrt{s-1}/s$ for $1 \le s \le 2$, where $F(1) = 2(e^\gamma/\pi)^{1/2} = 1.505\ldots$.*

Proof. (i) For $1 \le s \le 3$, $f \ge 0$ by the formula, and for $s > 1$, $\sqrt{s}f(s)$ is increasing because $F > 0$; so $f \ge 0$. For any g satisfying (5) with $\kappa = 1/2$ (for instance $g(\ell) = 1/\ell$ on the primes $\ell \equiv 3 \pmod 4$), the lower- and upper-bound properties of $\lambda^\pm$ give $V^-(D,z) \le V(z) \le V^+(D,z)$: indeed $\sum_d \lambda(d)g(d) = \sum_{n|P(z)} w(n)\sum_{d|n}\lambda(d)$ with $w(n) = g(n)\prod_{\ell|P(z)/n}(1-g(\ell)) \ge 0$. Letting $D \to \infty$ with s fixed in Cited Theorem 4.6 gives $f(s) \le 1 \le F(s)$. (ii) The formula is decreasing on $(0,2]$, and for $s > 2$, $F'(s) = (f(s-1) - F(s))/(2s) \le 0$ by (i). (iii) Put $\Delta = F - f \ge 0$. For $s > 2$, $(\sqrt{s}\Delta(s))' = -\Delta(s-1)/(2\sqrt{s}) \le 0$, so $\sqrt{s}\Delta(s) \le \sqrt{2}\Delta(2) \le \sqrt{2}F(2) = 2(e^\gamma/\pi)^{1/2} < 1.51$. (iv) $f(s) = F(1)s^{-1/2}\operatorname{arccosh}\sqrt{s}$, and $\operatorname{arccosh} y \ge \sqrt{y^2-1}/y$ for $y \ge 1$ (with $y = \cosh\tau$ this is $\tanh\tau \le \tau$). □

We also use Mertens' theorem in the progressions modulo 4: for $a \in \{1,3\}$, $\sum_{\ell\le y,\, \ell\equiv a\,(4)} 1/\ell = \frac{1}{2}\log\log y + b_a + O(1/\log y)$, which follows from the prime number theorem for these progressions. In particular

$$W(z) := \prod_{q\in\mathcal{Q},\, q<z}\Big(1-\frac{1}{q}\Big) \asymp (\log z)^{-1/2}. \tag{7}$$

4.3. **An upper bound.**

Proposition 4.8. $\#\{p \le x :\ p \equiv 1 \pmod 8,\ \mathrm{HC}(p) > 4\} \ll x/(\log x)^2$.

Proof. Let $\mathcal{A} = \{n \le x :\ n \equiv 1 \pmod 8\}$, $\mathcal{A}_2 = \emptyset$, and for odd ℓ let $\mathcal{A}_\ell$ be the set of $n \in \mathcal{A}$ with $n \equiv 0$, or (when $\ell \in \mathcal{Q}$) $n \equiv -4$ or $n \equiv -\bar{4} \pmod\ell$, where $4\bar{4} \equiv 1$. The number $\omega(\ell)$ of classes is 1 for $\ell \equiv 1 \pmod 4$, 2 for $\ell = 3$ (as $-4 \equiv -\bar{4} \pmod 3$) and 3 for $\ell \in \mathcal{Q}$, $\ell \ge 7$ ($-4 \equiv -\bar{4}$ only for $\ell \mid 15$). With $X = x/8$ and $g(\ell) = \omega(\ell)/\ell$ we have $|r_d| \le \omega(d) \le 3^{\omega(d)}$, and (4) holds with $\kappa = 2$ by Mertens' theorem modulo 4. Cited Theorem 4.5 with $z = x^{1/4}$ gives $S(\mathcal{A},z) \ll xV(z) + \sum_{d\le z} 3^{\omega(d)} \ll x(\log x)^{-2} + z(\log z)^2$. Every prime $p \in (z,x]$ with $p \equiv 1 \pmod 8$ and $\mathrm{HC}(p) > 4$ is counted: $\ell \nmid p$ for $\ell \le z$, and $p+4$, $4p+1$ have no prime factor in $\mathcal{Q}$ by Lemma 4.1. □

4.4. **The conditional lower bound.**

Theorem E. *There is an absolute constant $\eta_0 \in (0, 1/10]$ such that, if* $\mathrm{EH}(1-\eta_0)$ *holds, then for $x \ge x_0$*

$$\#\{p \le x : \ p \equiv 1 \pmod 8,\ p+4 \in \mathcal{S},\ 4p+1 \in \mathcal{S}\} \gg \frac{x}{(\log x)^2}.$$

In particular, under $\mathrm{EH}(1-\eta_0)$, $\mathrm{HC}(p) \ge 7$ *for $\gg x/(\log x)^2$ primes $p \le x$.*

Proof. Let $\eta \in (0, 1/10]$, to be fixed at the end as an absolute constant, and assume $\mathrm{EH}(1-\eta)$. Constants $c_1, C_2, \dots$ are absolute, and $x \ge x_0(\eta)$. Put $z = x^{1/2-\eta}$, $D = x^{1-\eta}/8$, $X = \mathrm{li}(x)/4$.

Step 1 (setup). Let $\mathcal{A}$ consist of the primes $p \le x$, $p \equiv 1 \pmod 8$, and let $\mathcal{A}_d$ (d squarefree, composed of primes of $\mathcal{Q}$) be the set of those p with $d \mid (p+4)(4p+1)$. For $q \in \mathcal{Q}$ the roots of $(b+4)(4b+1) \equiv 0 \pmod q$ are -4 and $-\bar{4}$; they are units and coincide only for $q \mid 15$. So the number of roots ν is multiplicative with $\nu(3) = 1$ and $\nu(q) = 2$ for $q \in \mathcal{Q}$, $q \ge 7$. By the Chinese remainder theorem $\#\mathcal{A}_d = \sum \pi(x; 8d, c)$ over $\nu(d)$ classes c modulo $8d$, so $\#\mathcal{A}_d = Xg(d) + r_d$ with $g(d) = \nu(d)/\varphi(d)$ and $|r_d| \le 2^{\omega(d)} E(x; 8d)$. The function g satisfies (5) with $\mathcal{P} = \mathcal{Q}$ and $\kappa = 1$: $g(3) = 1/2$, $g(q) = 2/(q-1)$, and $\sum_{q \in \mathcal{Q}, w \le q < z} 2/q = \log(\log z/\log w) + O(1/\log w)$.

Step 2 (lower bound). By (6) and Cited Theorem 4.6 with $\kappa = 1$,

$$S(\mathcal{A}, \mathcal{Q}, z) \ge XV(z)(f_1(s) - O((\log x)^{-1/6})) - \sum_{d \le D} \mu^2(d) 2^{\omega(d)} E(x; 8d),$$

where $s = \log D/\log z$. The moduli are $8d \le x^{1-\eta}$, so by $\mathrm{EH}(1-\eta)$ and Lemma 4.4 the last sum is $\ll x(\log x)^{-3}$. Next $s = \frac{1-\eta}{1/2-\eta} - O(\frac{1}{\log x}) = 2 + \frac{2\eta}{1-2\eta} - O(\frac{1}{\log x}) \in [2+\eta, 2.25]$, and $\log(1+u) \ge u/2$ on $[0,1]$ gives $f_1(s) \ge e^\gamma(s-2)/s \ge 0.79\,\eta$. Since $V(z) = \frac{1}{2}\prod_{q \in \mathcal{Q}, 7 \le q < z}(1 - \frac{2}{q-1}) \gg 1/\log z$ and $X \ge x/(5\log x)$, we get $S(\mathcal{A}, \mathcal{Q}, z) \ge c_1 \eta\, x/(\log x)^2$.

Step 3 (structure of the survivors). Let p be counted and $N \in \{p+4, 4p+1\}$. Then $N \equiv 5 \pmod 8$. An odd $N \equiv 1 \pmod 4$ has an even number of prime factors in $\mathcal{Q}$, counted with multiplicity, and here all of them are $\ge z$. As $N \le 4x+1 < z^3$, either $N \in \mathcal{S}$ or $N = q_1 q_2 m$ with $q_1 \le q_2$ in $\mathcal{Q}$, $q_1 \ge z$, $m \in \mathcal{S}$ and $m \le (4x+1)/z^2 \le 5x^{2\eta}$. Let B^{e}, B^{l} be the sets of counted p for which $p+4$, resp. $4p+1$, has the second shape. Then the left side of the theorem is at least $S(\mathcal{A}, \mathcal{Q}, z) - |B^{\mathrm{e}}| - |B^{\mathrm{l}}|$.

Step 4 (the bad sets). Let $p \in B^{\mathrm{e}}$, so $p + 4 = mq_1k$ with $k = q_2$. Fix m and q_1, put $a = mq_1$ and $Y = (x+4)/a$; then $q_1 \le k \le Y$, so only pairs with $q_1 \le \sqrt{(x+4)/m}$ occur, and $Y \ge z$. Let $w = x^{1/100}$ and sift the integers $k \in [1, Y]$ by the following classes, for primes $\ell \le w$: $k \equiv 0 \pmod \ell$ for every ℓ; $k \equiv 4\bar{a} \pmod \ell$ for odd $\ell \nmid a$; and $k \equiv 15\,\overline{4a} \pmod \ell$ for $\ell \in \mathcal{Q}$, $\ell \ge 7$, $\ell \nmid a$. The integer k coming from p avoids all of them: k is a prime $\ge z > w$; $ak - 4 = p$ is a prime $> w$; and $4ak - 15 = 4p + 1$ has no prime factor in $\mathcal{Q}$ below z. The three classes are distinct ($15 \not\equiv 0$ and $16 \not\equiv 15 \pmod \ell$ for $\ell \ge 7$), and $3 \nmid a$. So the number $\omega(\ell)$ of classes is at most $\omega^*(\ell)$,

where $\omega^*(2)=1$, $\omega^*(\ell)=3$ for $\ell\in\mathcal{Q}$, $\ell\ge 7$, and $\omega^*(\ell)=2$ otherwise; and $\omega(\ell)=1<\omega^*(\ell)$ exactly for the odd $\ell\le w$ dividing a, i.e. dividing m (as $q_1>w$). With $X=Y$ and $g(\ell)=\omega(\ell)/\ell$, we have $|r_d|\le\prod_{\ell\mid d}\omega(\ell)\le 3^{\omega(d)}$, and (4) holds with $\kappa=5/2$ and an absolute B (bound g by ω^*/ℓ); here $|r_d|\le\prod_{\ell\mid d}\omega(\ell)$. Cited Theorem 4.5 gives

$$\#\{p\in B^{\mathrm{e}}\text{ with these }m,q_1\}\le C\,Y\prod_{\ell\le w}\Big(1-\frac{\omega^*(\ell)}{\ell}\Big)\,h(m)+\sum_{d\le w}3^{\omega(d)}\le C_3\frac{Y\,h(m)}{(\log x)^{5/2}}+C\,w(\log w)^2,$$

where $h(m)=\prod_{\ell\mid m}\frac{\ell-1}{\ell-2}$ accounts for $(1-\frac{1}{\ell})/(1-\frac{2}{\ell})$ at $\ell\mid m$. The pairs (m,q_1) number $\ll\sum_{m\le 5x^{2\eta}}\sqrt{x/m}\ll x^{1/2+\eta}$, so the second term contributes $\ll x^{0.61+o(1)}$. For the first term,

$$\sum_{\substack{q\in\mathcal{Q}\\ z\le q\le\sqrt{2x}}}\frac{1}{q}=\frac12\log\frac{1/2}{1/2-\eta}+O\Big(\frac{1}{\log x}\Big)\le 2\eta,\qquad \sum_{\substack{m\in\mathcal{S}\\ m\le 5x^{2\eta}}}\frac{h(m)}{m}\le\prod_{\substack{\ell\equiv 1\,(4)\\ \ell\le 5x^{2\eta}}}\Big(1+\frac{1}{\ell-2}\Big)\le C_4(\eta\log x)^{1/2},$$

using $\sum_{j\ge1}h(\ell^j)\ell^{-j}=1/(\ell-2)$. Hence $|B^{\mathrm{e}}|\le C_2\eta^{3/2}x/(\log x)^2$. For B^{l} write $4p+1=mq_1k$, $Y=(4x+1)/(mq_1)$, $p=(ak-1)/4$, $p+4=(ak+15)/4$, and sift $k\equiv 0$, $k\equiv\bar a$ (odd $\ell\nmid a$) and $k\equiv -15\bar a$ ($\ell\in\mathcal{Q}$, $\ell\ge7$, $\ell\nmid a$); the same argument gives the same bound.

Step 5. The count is at least $(c_1\eta-2C_2\eta^{3/2})x/(\log x)^2-o(x/(\log x)^2)$. Take $\eta_0=\min(1/10,(c_1/4C_2)^2)$ and $\eta=\eta_0$. □

Remark 4.9. The proof follows the template of [5, §6]: a linear sieve on $(p+4)(4p+1)$ up to $x^{1/2-\eta}$, plus the observation that an integer $\equiv 1\pmod 4$ with no small prime factor in $\mathcal{Q}$ has either none or two large ones; the two-large-prime configurations cost $O(\eta^{3/2})$ against a main term of size η. The constant η_0 is not computed. With the Bombieri–Vinogradov theorem in place of EH, Steps 1–2 with $z=x^\zeta$, $0<\zeta<1/4$, and $D=x^{1/2}(\log x)^{-B}$ show only that $\gg_\zeta x/(\log x)^2$ primes $p\equiv1\pmod 8$ have $(p+4)(4p+1)$ free of prime factors in $\mathcal{Q}$ below x^ζ (here $s\to1/(2\zeta)>2$ and $f_1(s)>0$).

4.5. **An unconditional half.**

Theorem U. *There are absolute constants $\delta\in(0,1/100]$ and $c>0$ such that for $x\ge x_0$*

$$\#\{p\le x:\ p\equiv1\pmod 8,\ p+4\in\mathcal{S},\ 4p+1\text{ has no prime factor in }\mathcal{Q}\text{ below }x^\delta\}\ge c\,\frac{x}{(\log x)^2}.$$

For these p, $\Omega_{\mathcal{Q}}(4p+1)\le\log(4x+1)/(\delta\log x)<1/\delta+1$, and $\Omega_{\mathcal{Q}}(4p+1)$ is even.

The proof uses the vector sieve of Nath and Xie, which we quote in the case needed. Let $\mathcal{W}$ be a finite multiset of pairs of positive integers, $\mathcal{P}_1,\mathcal{P}_2$ sets of primes, $z_1,z_2\ge2$, $\mathbf{z}=(z_1,z_2)$, and $S(\mathcal{W};\mathbf{z})=\#\{(m,n)\in\mathcal{W}:(m,P_1(z_1))=(n,P_2(z_2))=1\}$. For $\mathbf{d}=(d_1,d_2)$ let $\mathcal{W}_{\mathbf{d}}=\{(m,n)\in\mathcal{W}:d_1\mid m,\ d_2\mid n\}$.

Cited theorem 4.10 (Vector sieve; [11, Prop. 3.10])**.** Suppose that $\#\mathcal{W}_{\mathbf{d}} = h(\mathbf{d})X + r(\mathbf{d})$ with h multiplicative on pairs, that for all primes ℓ

$$h(\ell,1)+h(1,\ell)-1 < h(\ell,\ell) \le h(\ell,1)+h(1,\ell), \quad h(\ell,1), h(1,\ell) \le \frac{C_1}{\ell}, \quad h(\ell,\ell) \le \frac{C_1}{\ell^2}, \tag{8}$$

and that $h_1(d) = h(d,1)$ and $h_2(d) = h(1,d)$ satisfy (5) on $\mathcal{P}_1, \mathcal{P}_2$ with dimensions κ_1, κ_2. Let $\log z_1 \asymp \log z_2$, $z_0 = \exp((\log z_1 z_2)^{1/3})$, $\sigma_i = \log D/\log z_i \ll 1$. Then for every $\varepsilon' > 0$

$$S(\mathcal{W};\mathbf{z}) \ge XV(z_0,h^*)V_1(z_1,z_0)V_2(z_2,z_0)\,\mathfrak{f}(\sigma_1,\sigma_2)(1+O((\log D)^{-1/6}))-O_{\varepsilon'}\Big(\sum_{\substack{d_1d_2\le D^{1+\varepsilon'}\\ d_i\mid P_i(z_i)}}\tau(d_1d_2)^4|r(d_1,d_2$$

where $V_i(z_i,z_0) = \prod_{\ell\in\mathcal{P}_i,\, z_0\le\ell<z_i}(1-h_i(\ell))$, $V(z_0,h^*) = \prod_{\ell<z_0}(1-h^*(\ell))$ with $h^*(\ell) = h(\ell,1)+h(1,\ell)-h(\ell,\ell)$ for $\ell \in \mathcal{P}_1\cap\mathcal{P}_2$ (in general [11, (3.10)]), and

$$\mathfrak{f}(\sigma_1,\sigma_2) = \sup\Big\{ f_{\kappa_1}(s_1)F_{\kappa_2}(s_2)+f_{\kappa_2}(s_2)F_{\kappa_1}(s_1)-F_{\kappa_1}(s_1)F_{\kappa_2}(s_2) :\ \frac{s_1}{\sigma_1}+\frac{s_2}{\sigma_2} = 1,\ s_i \ge \beta_{\kappa_i}\Big\}.$$

[11] state h with values in $(0,1]$; their own application [11, (4.3)] uses $h = 0$ on incompatible pairs, and the proof uses only $h \ge 0$. We use it the same way.

Proof of Theorem U. Let $\eta \in (0, 1/4]$, put $\theta_1 = (1-\eta)/2 \in [3/8, 1/2)$, $S_0 = \lceil 22801/\eta\rceil$, $\delta = \eta/(8S_0)$, $\varepsilon = \eta/16$, and $z_1 = x^{\theta_1}$, $z_2 = x^{\delta}$, $D = x^{1/2-2\varepsilon}$, $X = \mathrm{li}(x)/4$. The constants $K_1, K_2, \dots$ are absolute; the final choice of η is absolute, and $x \ge x_0(\eta)$.

Step 1 (parity). Let $\mathcal{W} = \{(p+4, 4p+1) :\ p \le x,\ p \equiv 1\ (8)\}$, $\mathcal{P}_1 = \mathcal{P}_2 = \mathcal{Q}$, and $S = S(\mathcal{W};(z_1,z_2))$. If p is counted by S, then $p+4$ is $\equiv 1 \pmod 4$ and its prime factors in $\mathcal{Q}$ are $\ge z_1$, even in number, and fewer than four since $z_1^4 \ge x^{3/2} > x+4$. So either $p+4 \in \mathcal{S}$, or $p+4 = mqq'$ with $z_1 \le q \le q'$ in $\mathcal{Q}$ and $m \in \mathcal{S}$. Hence the left side of the theorem is at least $S - T_1$, where T_1 counts the p of the second kind.

Step 2 (main term). For squarefree d_1, d_2 composed of primes of $\mathcal{Q}$, the conditions $d_1 \mid p+4$, $d_2 \mid 4p+1$ fix p modulo each prime of $l = \mathrm{lcm}(d_1,d_2)$: modulo $q \mid d_1$ as -4, modulo $q \mid d_2$ as $-\bar{4}$, and modulo 3 the two conditions agree ($p \equiv 2$). A prime $q \in \mathcal{Q}$, $q \ge 7$, divides at most one of $p+4$ and $4p+1$, since $(4p+1)-4(p+4) = -15$. So $\#\mathcal{W}_{\mathbf{d}}$ is 0 if some $q \ge 7$ divides $\gcd(d_1,d_2)$, and otherwise equals $\pi(x;8l,c)$ for one unit class c. Define $h(\mathbf{d}) = 1/\varphi(l)$ in the latter case and $h(\mathbf{d}) = 0$ in the former, and $h = 0$ on pairs involving primes outside $\mathcal{Q}$. Then $|r(\mathbf{d})| \le E(x;8l)$ and (8) holds with $C_1 = 5$: $h(q,1) = h(1,q) = 1/(q-1)$, $h(q,q) = 0$ for $q \ge 7$, and $h(3,1) = h(1,3) = h(3,3) = 1/2$. Each h_i satisfies (5) with $\kappa_i = 1/2$ (Mertens modulo 4). A squarefree l is the lcm of at most $3^{\omega(l)}$ pairs, and $\tau(d_1d_2) \le \tau(l)^2$. With $\varepsilon' = \varepsilon$ we have $8D^{1+\varepsilon} \le x^{1/2-\varepsilon}$, so by Bombieri–Vinogradov and Lemma 4.4 the remainder in Cited Theorem 4.10 is $\ll x(\log x)^{-4}$.

The product: $h^*(3)=1/2$ and $h^*(q)=2/(q-1)$ for $q\ge 7$, so

$$\frac{V(z_0,h^*)V_1(z_1,z_0)V_2(z_2,z_0)}{W(z_1)W(z_2)}=\frac{9}{8}\prod_{\substack{q\in\mathcal{Q}\\7\le q<z_0}}\frac{q^2(q-3)}{(q-1)^3}\prod_{i=1,2}\prod_{\substack{q\in\mathcal{Q}\\z_0\le q<z_i}}\Big(1-\frac{1}{(q-1)^2}\Big)\to\mathfrak{G}:=\frac{9}{8}\prod_{\substack{q\in\mathcal{Q}\\q\ge 7}}\frac{q^2(q-3)}{(q-1)^3}>$$

In $\mathfrak{f}(\sigma_1,\sigma_2)$ take $s_2=S_0$ and $s_1=(1/2-2\varepsilon-S_0\delta)/\theta_1$; then $s_1\theta_1+s_2\delta=1/2-2\varepsilon$ as required, and $s_1=\frac{1-\eta/2}{1-\eta}\in[1+\eta/2,\,7/6]$. Write $F=F_{1/2}$, $f=f_{1/2}$. By Lemma 4.7,

$$f(s_1)F(S_0)+F(s_1)f(S_0)-F(s_1)F(S_0)=f(s_1)F(S_0)-F(s_1)(F-f)(S_0)\ge f(s_1)-F(1)\cdot\frac{1.51}{\sqrt{S_0}},$$

and $f(s_1)\ge F(1)\sqrt{\eta/2}/(7/6)\ge 0.6F(1)\eta^{1/2}$ while $1.51/\sqrt{S_0}\le\eta^{1/2}/100$. Therefore

$$S\ge(\tfrac12\mathfrak{G}F(1)\eta^{1/2}-o(1))XW(z_1)W(z_2).\tag{9}$$

Step 3 (the switched term). Let p be counted by T_1, with $p+4=mqq'$. Fix m and q, put $a=mq$ and $Y=(x+4)/a$. Then $q\le q'\le Y$, so $q\le\sqrt{(x+4)/m}$, and $m\le(x+4)/z_1^2$. Let $w=x^{1/16}$; note $z_2<w<z_1$. Sift the integers $k\in[1,Y]$ at the primes $\ell\le w$ by: $k\equiv 0$ for all ℓ; $k\equiv 4\bar a$ for odd $\ell\nmid a$; and $k\equiv 15\,\overline{4a}$ for $\ell\in\mathcal{Q}$, $7\le\ell<z_2$, $\ell\nmid a$. The integer $k=q'$ avoids all these classes: $q'\ge z_1>w$ is prime, $aq'-4=p\ge z_1^2-4>w$ is prime, and $4aq'-15=4p+1$ has no prime factor in $\mathcal{Q}$ below z_2. As in Step 4 of Theorem E the number of classes is at most $\omega^*(\ell)$ (with $\omega^*=3$ on $\mathcal{Q}\cap[7,\infty)$, 1 at 2, 2 otherwise), so (4) holds with $\kappa=5/2$ and an absolute B, uniformly in m,q,η; again $|r_d|\le 3^{\omega(d)}$. The odd primes $\ell\le w$ dividing a divide m, so they are $\equiv 1\pmod 4$; there only the class 0 is removed. Hence

$$V(w)=\frac12\prod_{3\le\ell\le w}\Big(1-\frac{2}{\ell}\Big)\prod_{\substack{\ell\mid m\\\ell\le w}}\frac{\ell-1}{\ell-2}\prod_{\substack{\ell\in\mathcal{Q}\\7\le\ell<z_2}}\Big(1-\frac{1}{\ell-2}\Big)\le K_1\frac{h(m)\,W(z_2)}{(\log x)^2},$$

with h as in the proof of Theorem E. Cited Theorem 4.5 gives $\#\{p\text{ in }T_1\text{ with these }m,q\}\le K_2'\,Y\,h(m)W(z_2)(\log x)^{-2}+O(w(\log w)^2)$. The pairs (m,q) number $\ll x^{1/2+\eta/2}$, so the second terms add up to $O(x^{0.7})$. For the first terms let $L=\log x$ and $\beta=\log m/L$. Mertens' theorem modulo 4 and $\log\rho\le\rho-1$ give

$$\sum_{\substack{q\in\mathcal{Q}\\z_1\le q\le\sqrt{(x+4)/m}}}\frac1q\le\frac12\log\frac{(1-\beta)L+1}{2\theta_1L}+\frac{K_3}{L}\le\frac23(\eta-\beta)_+\,+\frac{K_4}{L},$$

because $\beta\le\eta+O(1/x)$ for $m\le(x+4)/z_1^2$ and $1/(2(1-\eta))\le 2/3$. Let $H(t)=\sum_{m\in\mathcal{S},m\le t}h(m)/m\le\prod_{\ell\equiv1\,(4),\,\ell\le t}(1+\frac{1}{\ell-2})\le K_5(\log 3t)^{1/2}$. Partial summation gives

$$\sum_{\substack{m\in\mathcal{S}\\m\le e^{\eta L}}}\frac{h(m)}{m}\Big(\eta-\frac{\log m}{L}\Big)=\frac1L\int_1^{e^{\eta L}}H(t)\frac{dt}{t}\le\frac{K_5}{L}\int_0^{\eta L}(\log 3+u)^{1/2}du\le 2K_5\,\eta^{3/2}L^{1/2}$$

for $\eta L \ge 2$, while $\frac{K_4}{L} H(2x) = O(L^{-1/2})$. Altogether

$$T_1 \le K_6\, \eta^{3/2} \frac{x\, W(z_2)}{(\log x)^{3/2}} + o\Big(\frac{x\, W(z_2)}{(\log x)^{3/2}}\Big) \le (K_7 \eta^{3/2} + o(1)) X W(z_1) W(z_2),$$

because $XW(z_1) \gg x(\log x)^{-3/2}$ for $\theta_1 \ge 3/8$, with an absolute implied constant.

Step 4 (choice of η). By (9) and Step 3 the left side of the theorem is at least $XW(z_1)W(z_2)\eta^{1/2}(\frac{1}{2}\mathfrak{G}F(1) - K_7\eta - o(1))$. Take $\eta = \min(1/4, \mathfrak{G}F(1)/(4K_7))$. Then S_0, δ, ε are absolute, $\delta \le 1/100$, and $XW(z_1)W(z_2) \asymp_\delta x/(\log x)^2$. The bound on $\Omega_{\mathcal{Q}}$ holds because every prime factor of $4p+1 \le 4x+1$ in $\mathcal{Q}$ is $\ge x^\delta$, and $\Omega_{\mathcal{Q}}$ is even because $4p + 1 \equiv 1 \pmod 4$. □

Remark 4.11. (i) The mechanism is Iwaniec's: the semi-linear lower-bound function grows like $\sqrt{s-1}$ beyond its sifting limit $s = 1$, while the configurations with two large prime factors in $\mathcal{Q}$ have mass $O(\eta^{3/2})$; the second component is sifted only to x^δ at the cost of a level $x^{S_0\delta} = x^{\eta/8}$. (ii) The same proof with the roles of $p+4$ and $4p+1$ exchanged gives $\gg x/(\log x)^2$ primes $p \equiv 1 \pmod 8$ with $4p+1 \in \mathcal{S}$ and $p+4$ free of primes of $\mathcal{Q}$ below x^δ: in Step 3 write $4p+1 = mqq'$, $p = (aq'-1)/4$, and sift $q' \equiv \bar{a}$ and $q' \equiv -15\bar{a}$. (iii) The proof gives no useful numerical value of δ. Numerically, among the 22 280 854 primes $p \le 10^{10}$ with $p \equiv 1 \pmod 8$ and $p+4 \in \mathcal{S}$, the value $\Omega_{\mathcal{Q}}(4p+1)$ is $0, 2, 4, 6, 8, 10$ for 7 962 940, 12 902 238, 1 378 173, 37 005, 496 and 2 of them.

Remark 4.12 (What is missing for (2))**.** Theorem U proves one shift exactly; the other must lose its boundedly many prime factors from $\mathcal{Q}$. In the implementations above this needs primes in the residue classes of $(p+4)(4p+1)$, which vary with the modulus, at a level of distribution close to 1: Theorem E assumes EH$(1-\eta_0)$, and a semi-linear⊗semi-linear vector sieve without switching needs level $\cosh^2(1/2) > 1$, as one checks from $f_{1/2}/F_{1/2} = \operatorname{arccosh}\sqrt{s}$ on $[1,2]$. These are statements about particular sieve set-ups, not barriers (the threshold $\cosh^2(1/2)$ is computed for the symmetric choice $z_1 = z_2 = x^{1/2}$, $s_1 = s_2$, which is optimal among the choices with $s_1, s_2 \in [1,2]$ by concavity of $\operatorname{arccosh}\sqrt{s}$); Friedlander and Iwaniec's determinant-1 problem (Remark 4.2) has the same conditional status.

5. Dickson-type constructions and their reach

Dickson's conjecture [2] asserts: if $a_i s + b_i$ $(1 \le i \le k)$ are linear forms with integers $a_i \ge 1$, b_i, and no prime divides $\prod_i (a_i s + b_i)$ for every s, then for infinitely many $s \in \mathbb{N}$ all $a_i s + b_i$ are prime.

Proposition 5.1. *Assume Dickson's conjecture. For every $X \ge 4$ there are infinitely many primes $p \equiv 1 \pmod{24}$ with $\mathrm{HC}(p) > X$.*

Proof. Let $\mathcal{C}_X = \{x \in [4, X] :\ 4 \mid x \text{ or } x \equiv 7\ (8)\}$ and $Q = \operatorname{lcm}(24,\ x(x+1) : x \in \mathcal{C}_X)$. Consider the forms $p(s) = 1 + Qs$, $F_x(s) = 1 + Qs/(x+1)$ and

$G_x(s) = 1 + xQs/(x+1)$ for $x \in \mathcal{C}_X$, so that $p + x = (x+1)F_x$ and $xp + 1 = (x+1)G_x$. All forms take the value 1 at $s = 0$, hence have no fixed prime divisor. Let s be large with all forms prime. Then $p \equiv 1 \pmod{24}$, so $\mathcal{C}_p \cap [1, X] = \mathcal{C}_X$. Let $x \in \mathcal{C}_X$. As $x(x+1) \mid Q$, we have $F_x \equiv G_x \equiv 1 \pmod{x}$, and $F_x, G_x > x + 1$ are primes. The divisors of $p + x$ are e and eF_x with $e \mid x + 1$, and $eF_x \equiv e \pmod{x}$; a divisor $\equiv -1 \pmod{x}$ would give $e \mid x + 1$ with $e \equiv -1 \pmod{x}$, i.e. $e = x - 1$ (impossible for $x \geq 4$, as $x - 1 \nmid x + 1$) or $e \geq 2x - 1 > x + 1$. The same holds for $xp + 1$. So no certificate has $c \leq X$. □

The primes produced are $\equiv 1$ modulo every prime $\ell \leq X + 1$, so $\log p \geq (1 + o(1))X$. The next theorem shows that this is forced for any construction of this shape.

Theorem W. *Let $X \geq 8$, let $P \in \mathbb{Z}[s]$ be nonconstant and let $\mathcal{V} \subseteq \mathbb{N}$ be infinite, such that $P(s)$ is a prime $\equiv 1 \pmod 8$ exceeding X for every $s \in \mathcal{V}$. Suppose that for every $x \in 4\mathbb{N} \cap [4, X]$ there is an identity $P(s) + x = c_x \prod_i f_{x,i}(s)^{e_{x,i}}$ in $\mathbb{Z}[s]$, with $c_x \in \mathbb{Z} \setminus \{0\}$ and nonconstant $f_{x,i}$, and nonzero integers $C_{x,i}$ such that $f_{x,i}(s)/C_{x,i}$ is prime for every $s \in \mathcal{V}$. Then for every odd prime $\ell \leq \lfloor X/4 \rfloor + 1$ the residue of $P(s)$ modulo ℓ is the same for all $s \in \mathcal{V}$. Consequently every element of $P(\mathcal{V})$ except possibly the least satisfies $\log P(s) > \sum_{3 \leq \ell \leq \lfloor X/4 \rfloor + 1} \log \ell = (\frac{1}{4} + o(1))X$.*

Proof. Fix an odd prime $\ell \leq \lfloor X/4 \rfloor + 1$; then $\ell \leq X$. A nonconstant polynomial takes each value finitely often, so we may choose $s \in \mathcal{V}$ with $|f_{x,i}(s)| > \ell |C_{x,i}|$ for all x, i. Let $p = P(s)$ and $a \equiv p \pmod{\ell}$, $1 \leq a \leq \ell - 1$ (as $p > X \geq \ell$). Choose $1 \leq k \leq \ell - 1$ with $4k \equiv -a \pmod{\ell}$ and put $x = 4k \leq X$. Then $\ell \mid p + x = c_x \prod_i f_{x,i}(s)^{e_{x,i}}$. If $\ell \mid f_{x,i}(s) = C_{x,i} r$ with r prime and $\ell \nmid C_{x,i}$, then $\ell = r$ and $|f_{x,i}(s)| = \ell |C_{x,i}|$, contrary to the choice of s. Hence $\ell \mid c_x$ or $\ell \mid C_{x,i}$ for some i. In either case ℓ divides $P(s') + x$ for every $s' \in \mathcal{V}$, i.e. $P(s') \equiv -x \equiv a \pmod{\ell}$. The elements of $P(\mathcal{V})$ therefore lie in one class modulo $\Pi = \prod_{3 \leq \ell \leq \lfloor X/4 \rfloor + 1} \ell$, so all but the least exceed Π; finally $\log \Pi = (1/4 + o(1))X$ by the prime number theorem. □

The family of Proposition 5.1 satisfies the hypotheses ($c_x = x + 1$, $f_{x,1} = F_x$). So a fixed polynomial family whose window integers factor as fixed constants times primes can certify $\mathrm{HC}(p) > X$ only when $X < (4 + o(1)) \log p$, apart from its least member. Theorem W bounds what such a family certifies, not the actual HC of its members, and it does not apply to sequences of families or to mechanisms like Theorem Q, where the window integers are not products of primes of a fixed shape.

6. Coverings fail on non-square classes

By Lemma 2.3 a certificate (c, d) of a prime p is a congruence condition on p: $p \equiv -c \pmod{d}$ (U-line) or $p \equiv -\bar{c} \pmod{d}$ (T-line), together with $c \in \mathcal{C}_p$, which is a condition modulo 8. Corollary 2.8 shows that a finite set

of such congruences cannot cover the primes in a square class. The following theorem gives classes without squares that cannot be covered either.

Theorem N. *Let Λ be a nonempty finite set of primes with $\ell_0 := \min\Lambda \geq 11$, and let R be an integer with $1 \leq R < (\ell_0 - 1)/3$. Let $C \geq 3$ and $Y \geq 2$. Put $\mathcal{T} = (\{\text{primes} \leq Y\} \setminus \Lambda) \cup \{2\}$; for $\ell' \in \mathcal{T}$ let $a(\ell')$ be the least integer with $\ell'^{a(\ell')} > C+1$, except that $a(2)$ is the maximum of 3 and this number; put $M = \prod_{\ell' \in \mathcal{T}} \ell'^{a(\ell')}$; and for $\ell \in \Lambda$ let J_ℓ be the least integer with $\ell^{J_\ell} > R(C+1)$. Then every prime $p > C$ with*

$$p \equiv 1 \pmod{M} \quad \text{and} \quad p \equiv R \pmod{\ell^{J_\ell}}\ (\ell \in \Lambda)$$

has no certificate (c, d) with $c \leq C$ and all prime factors of d in $\mathcal{T} \cup \Lambda$. There are infinitely many such primes, all $\equiv 1 \pmod 8$.

Proof. Infinitude follows from Dirichlet's theorem, since $\gcd(R, \ell) = 1$ for $\ell \in \Lambda$. Let p be such a prime; then $p \equiv 1 \pmod 8$ and $\mathcal{C}_p = 4\mathbb{N} \cup (7 + 8\mathbb{Z}_{\geq 0})$. Suppose (c, d) is a certificate with $c \leq C$, so $4 \leq c \leq C$, $d \equiv -1 \pmod c$, and $d \mid N$ where $N = p + c$ or $N = cp + 1$. Write $d = Ls$ with $L = \prod_{\ell \in \Lambda} \ell^{j_\ell}$ and s composed of primes of $\mathcal{T}$. Note $N \equiv c + 1 \pmod{\ell'^{a(\ell')}}$ for $\ell' \in \mathcal{T}$, since $p \equiv 1$ there.

Step 1: $s \mid c + 1$. Let $\ell' \mid s$. If $\ell'^{a(\ell')} \mid s$, then $\ell'^{a(\ell')} \mid N$, so $\ell'^{a(\ell')} \mid c + 1$, impossible as $0 < c + 1 < \ell'^{a(\ell')}$. So $s \mid M$, hence $N \equiv c + 1 \pmod s$ and $s \mid c + 1$. Put $m = (c + 1)/s \in \mathbb{N}$.

Step 2: $L \geq \ell_0$. By Lemma 2.7, $(d/p) = -1$. For odd $\ell' \in \mathcal{T}$, $(\ell'/p) = (p/\ell') = (1/\ell') = 1$, and $(2/p) = 1$; so $(s/p) = 1$ and $L \neq 1$.

Step 3: $L \equiv -m \pmod c$. From $Ls \equiv -1$ and $sm = c + 1 \equiv 1 \pmod c$: $L \equiv Lsm \equiv -m$.

Step 4: $c \equiv -R \pmod L$ on the U-line and $Rc \equiv -1 \pmod L$ on the T-line. Let $\ell \in \Lambda$ with $j = j_\ell \geq 1$ and $j_0 = \min(j, J_\ell)$. Then $\ell^{j_0} \mid N$ and $p \equiv R \pmod{\ell^{j_0}}$, so $c + R \equiv 0$ (U) or $Rc + 1 \equiv 0$ (T) modulo ℓ^{j_0}. If $j > J_\ell$, this gives $c + R \geq \ell^{J_\ell} > R(C + 1)$, so $c > RC \geq C$, or $Rc + 1 > R(C + 1)$, so $c > C$; both are impossible. Hence $j \leq J_\ell$ for all ℓ, and the Chinese remainder theorem gives the claim modulo L.

Step 5: the U-line is impossible. Write $c + R = (i + 1)L$ with $i \geq 0$. Modulo c, Step 3 gives $-(i + 1)m \equiv R$, so $(i + 1)m + R = kc$ with $k \in \mathbb{N}$. As $m \mid c + 1$, the number $kc - R = (i + 1)m$ divides $(i + 1)(c + 1)$; put $g = (i+1)(c+1)/(kc-R) \in \mathbb{N}$. Then $c(gk - i - 1) = i + 1 + gR > 0$, so $e := gk-i-1 \geq 1$ and $g \leq gk = e+i+1$. Hence $ce = i+1+gR \leq Re+(R+1)(i+1)$, i.e. $(c - R)e \leq (R + 1)(i + 1)$, which gives $c \leq R + (R + 1)(i + 1)$ (trivially if $c \leq R$). With $c = (i + 1)L - R$ this is $L \leq R + 1 + 2R/(i + 1) \leq 3R + 1 < \ell_0$, contradicting Step 2.

Step 6: the T-line is impossible. Write $Rc + 1 = nL$, $n \in \mathbb{N}$. Modulo c, $nL \equiv 1$ and Step 3 give $nm + 1 = kc$ with $k \in \mathbb{N}$. As $m \mid c + 1$, $kc - 1 = nm$ divides $n(c + 1)$; put $g = n(c+1)/(kc-1) \in \mathbb{N}$. Then $c(gk - n) = g + n > 0$, so $e := gk - n \geq 1$, $g \leq e + n$, and $ce \leq e + 2n$; hence $c \leq 2n + 1$. With

$c = (nL-1)/R$ this gives $L \le 2R + (R+1)/n \le 3R+1 < \ell_0$, again a contradiction. □

Corollary 6.1. *Let $\ell \ge 11$ be prime. Then there is a quadratic non-residue R modulo ℓ with $1 \le R < (\ell-1)/3$. For any such R, every $C \ge 3$ and every finite set $\mathcal{D} \subset \mathbb{N}$, there are infinitely many primes $p \equiv 1 \pmod{840}$, $p \equiv R \pmod{\ell}$, having no certificate (c,d) with $c \le C$ and $d \in \mathcal{D}$. In particular no finite set of certificate congruences covers the primes of the class $\{p \equiv 1 \pmod{840},\ p \equiv R \pmod{\ell}\}$, which contains no squares.*

Proof. For $\ell = 11, 13, 19$ take $R = 2$, and for $\ell = 17$ take $R = 3$. For $\ell \ge 23$ let n be the least quadratic non-residue and $m' = \lceil \ell/n \rceil$. Then $0 < m'n - \ell < n$, so $m'n - \ell$ is a residue and m' is a non-residue; thus $m' \ge n$, and $n \le m' < \ell/n + 1$ gives $n(n-1) < \ell$, so $n < \sqrt{\ell} + 1 \le (\ell-1)/3$. For the second claim apply Theorem N with $\Lambda = \{\ell\}$ and $Y \ge 7$ at least the largest prime factor of the elements of $\mathcal{D}$; then $2, 3, 5, 7 \in \mathcal{T}$ and $840 \mid M$. A square $\equiv R \pmod{\ell}$ would make R a residue. □

Remark 6.2. (i) Steps 3–6 use about the primes of Λ only $p \equiv R$; the non-residue condition is not needed for Theorem N, only for the conclusion that the class contains no squares. In a square class the Jacobi obstruction (Corollary 2.8) already excludes every certificate divisor supported on small primes; in the classes of Corollary 6.1 it does not, since $(\ell/p) = -1$, and the obstruction comes from the size conditions $d \equiv -1 \pmod{c}$, $d \mid N$. (ii) For $\Lambda = \{3\}, \{5\}, \{7\}$ no non-residue $R < (\ell-1)/3$ exists, and the equations of Steps 5–6 do have solutions, for example $(c,d) = (4,3)$, $(8,15)$, $(4,7)$; these are certificates in Mordell's covering modulo 840. (iii) Theorem N allows Λ to be large: with $R = 2$ and Λ the primes $\equiv \pm 3 \pmod 8$ in $[11, Y]$ one obtains primes that are non-residues modulo all these ℓ and still have no certificate with $c \le C$ supported on the primes $\le Y$. (iv) Consequently a bound $\mathrm{HC}(p) \le F(\ell_p)$ in terms of the least prime non-residue ℓ_p of p cannot be proved by a finite certificate family for each value of ℓ_p: already $\ell_p = 11$, $p \equiv 2 \pmod{11}$ is not coverable. Arguments whose moduli grow with p are not excluded.

7. Computations

Computation 7.1. For every prime $p \equiv 1 \pmod 4$ with $p < 10^{12}$ we have $\mathrm{HC}(p) \le 64$. There are 18 803 924 340 such primes; 9 401 972 490 of them are $\equiv 5 \pmod 8$ (where $\mathrm{HC} = 3$) and 9 401 951 850 are $\equiv 1 \pmod 8$. For the latter the distribution of HC is:

HC	count	HC	count	HC	count	HC	count
4	8 851 945 528	20	169 702	36	336	52	3
7	511 181 865	23	49 652	39	348	55	5
8	31 048 619	24	11 842	40	56	56	2
12	5 448 736	28	3 024	44	80	60	1
15	1 687 276	31	3 086	47	31	64	1
16	400 730	32	924	48	3		

The value 64 occurs only at $p = 21\,841$ (certificate $(64, 575)$: $575 \mid 64p + 1$). The largest value on $[10^8, 10^{12})$ is 60, at $p = 818\,789\,590\,249$.

Method. A segmented sieve enumerates the primes. For each $p \equiv 1 \pmod 8$ the program scans $c \in \mathcal{C}_p$ upwards from 4, factors $p + c$ and $cp + 1$ completely (trial division below 1000, Brent–Pollard ρ, and the Miller–Rabin test with the first twelve primes as bases, which is deterministic below $3.18 \cdot 10^{23}$ [14]; all numbers are below $4 \cdot 10^{15}$), enumerates all divisors, and stops at the first divisor $\equiv -1 \pmod c$. By Lemma 2.3 this computes $\mathrm{HC}(p)$ exactly.

Checks. (a) The same sieve counts $\pi(10^{12}) = 37\,607\,912\,018$, the known value, and $\pi(10^{12}; 4, 1)$ equals the total above. (b) Below 10^9 an independently written program gives the identical histogram. (c) All 126 primes below 10^{12} with $\mathrm{HC}(p) \geq 44$ were re-verified in Python (certificate valid and no smaller $c \in \mathcal{C}_p$ certified), and the 12 primes with $\mathrm{HC}(p) \geq 52$ were recomputed once more by the independent script `es-hc-note.check.py`. (d) For this note the program was rerun on $[0, 10^9)$ and on the last block $[9.52 \cdot 10^{11}, 10^{12})$; both outputs agree with the archived ones. The full range was not recomputed independently.

Remarks. (i) By Proposition 2.2 every prime $p \equiv 1 \pmod 4$ below 10^{12} has a solution of (1) built from a word with a coordinate at most 64; the Erdős–Straus conjecture itself is verified much further [12, 9]. (ii) The proportion of primes $p \equiv 1 \pmod 8$ with $\mathrm{HC}(p) > 4$ is 0.0868 below 10^8 and 0.0564 on $[9.52 \cdot 10^{11}, 10^{12})$; the number with $\mathrm{HC}(p) > 4$ below 10^{12} is $550\,006\,322 \approx 0.420\, x/(\log x)^2$. The counts below $10^5, 10^6, \ldots, 10^{11}$ are 295, 2 226, 16 125, 125 059, 986 400, 7 962 940, 65 618 311, with ratio to $x/(\log x)^2$ between 0.39 and 0.43, consistent with the order in Proposition 4.8 and Theorem E. (iii) All 5 719 primes $p = 80k^2 + 1 \leq 10^{12}$ (the primes of Theorem Q) were checked to have $\mathrm{HC}(p) \geq 7$, as the theorem predicts.

8. Open problems

Problem 1. Prove (2) unconditionally, i.e. $\mathrm{HC}(p) > 4$ for infinitely many primes p. By Theorem U it suffices to remove the boundedly many prime factors from $\mathcal{Q}$ of $4p + 1$.

Problem 2. Is HC unbounded? For every fixed $X \geq 7$ the only route we know is Dickson's conjecture (Proposition 5.1).

Problem 3. Prove $\mathrm{HC}(p) \le F(p)$ for all large primes p, for some function F. Any such bound proves the Erdős–Straus conjecture for all large primes. The data suggest that F could be a power of $\log p$; Theorem N shows that a proof cannot consist of a fixed finite family of certificate congruences, even away from the square classes.

Problem 4. Does every large prime have a certificate (c, d) with c and d bounded by a power of $\log p$? A positive answer would be a covering whose moduli grow polylogarithmically, which Theorem N does not exclude. On the other hand $d \le \kappa \log p$ does not suffice for a small absolute $\kappa > 0$: let $m = \prod_{\ell \le Y} \ell$; Linnik's theorem [7] gives a prime $p \equiv 1 \pmod{8m}$ with $\log p \ll Y$, and by Corollary 2.8(b) every certificate divisor of p has a prime factor $> Y$.

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